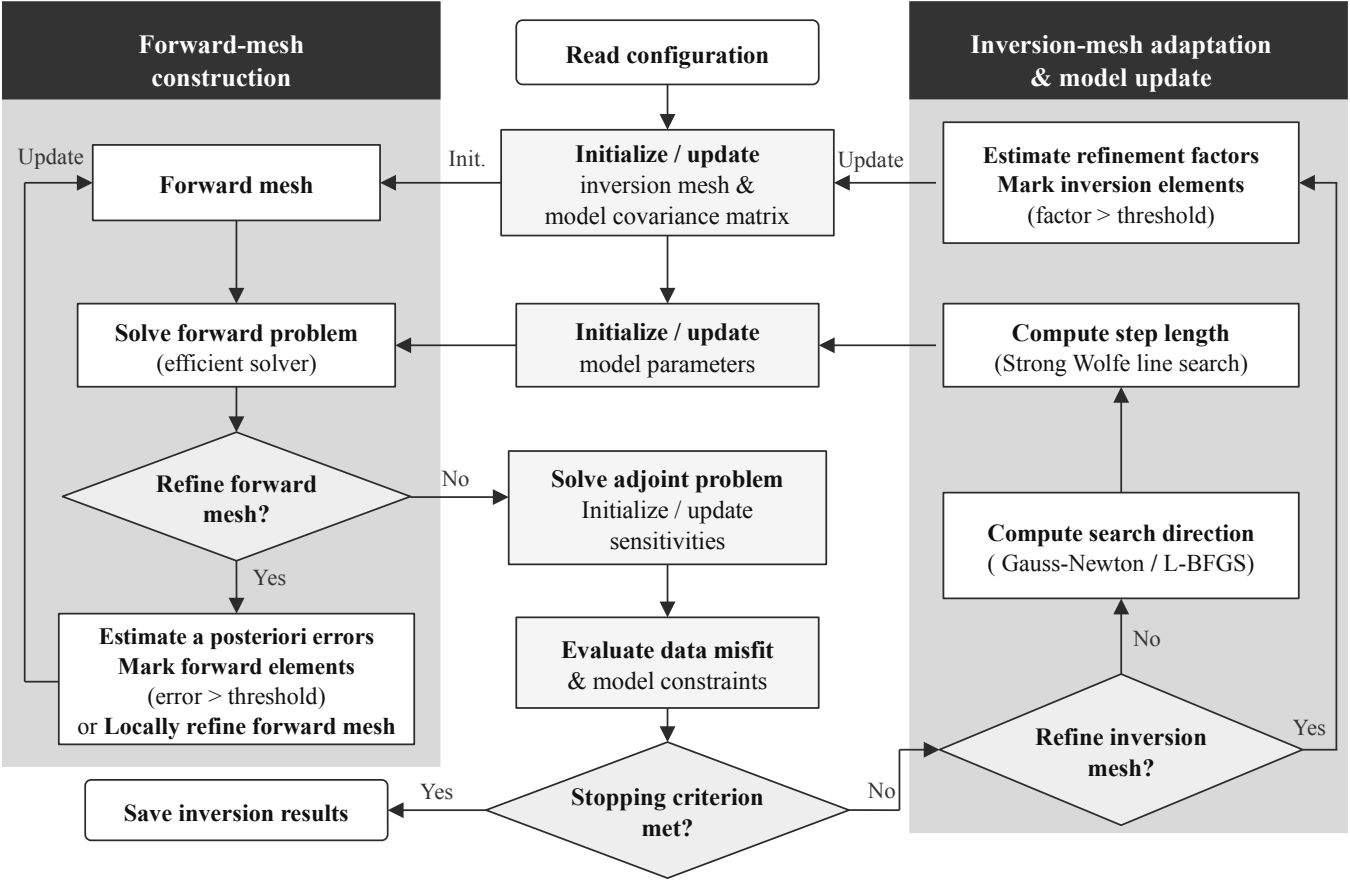
Forward-mesh construction
Read configuration
Inversion-mesh adaptation & model update
Update
Forward mesh
Init.
Initialize / update inversion mesh & model covariance matrix
Update
Estimate refinement factors
Mark inversion elements
(factor > threshold)
Solve forward problem
(efficient solver)
Initialize / update model parameters
Compute step length
(Strong Wolfe line search)
Refine forward mesh?
No
Solve adjoint problem
Initialize / update sensitivities
Compute search direction
(Gauss-Newton / L-BFGS)
Yes
Estimate a posteriori errors
Mark forward elements
(error > threshold)
or Locally refine forward mesh
Evaluate data misfit
& model constraints
No
Refine inversion mesh?
Yes
Save inversion results
Yes
Stopping criterion met?
No